Aug 12

Independence

Two events A & B are said to be independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

Ex: ① Throw a fair die $S = \{1, 2, 3, 4, 5, 6\}$

(a) $A = \{1, 2\}$ $B = \{2, 3\}$ dependent

(b) $A = \{1, 2\}$ $B = \{2, 3, 4\}$ independent

(c) $A = \{1, 2\}$ $B = \{3, 4\}$ dependent

② Draw a card.

$A = \{\text{card is spade}\}$ $B = \{\text{card is king}\}$

independent

③ Throw a die & toss a coin

$$S = \{H, T\} \times \{1, 2, \ldots, 6\}$$

A: toss is H B: die shows even no.

indep.

④ Throw a die n times independently.

A: event on first 5 throws B: event on throws 11–15.

indep.

Independence of 3 events A, B, C

Mutually independent:

$$P(A \cap B) = P(A) \cdot P(B)$$
$$P(B \cap C) = P(B) \cdot P(C)$$
$$P(A \cap C) = P(A) \cdot P(C)$$

pairwise independent.

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

Ex: $S = \{1, 2, \dots, 24\}$ $P(i) = \frac{1}{24}$

A $\{1, 4, \dots, 12\}$ B $\{2, 9, 6, \dots, 24\}$ C $\{7, 8, \dots, 18\}$

Ex: ① n_r red, n_b blue $\quad k \le n_r$, $k \le n_b$

Draw k marbles without replacement.

$p_k = P(k\text{-th draw is red}) = \frac{n_r}{n_r + n_b}$ $\longleftrightarrow$ with replacement

$p_1 = \frac{n_r}{n_r + n_b}$ $\quad p_2 = \frac{n_r}{n_r+n_b} \cdot \frac{n_r - 1}{n_r + n_b - 1} + \frac{n_b}{n_r+n_b} \cdot \frac{n_r}{n_r + n_b - 1}$

$= \frac{n_r}{n_r + n_b}$

② Throw a die repeatedly

(a) A: eventually a 5 occurs.

$A = \{5, \bar{5}5, \bar{5}\bar{5}5, \bar{5}\bar{5}\bar{5}5, \dots\}$

$P(A) = \frac{1}{6} + \frac{5}{6}\frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \dots = \frac{1}{6} \cdot \frac{1}{1 - 5/6} = 1$

$P(A) = \frac{1}{6} + \frac{5}{6} P(A)$

(b) B : 4 appears before 5

$B = \{4, \overline{4,5}\,4, \overline{4,5}^2 4, \dots\}$

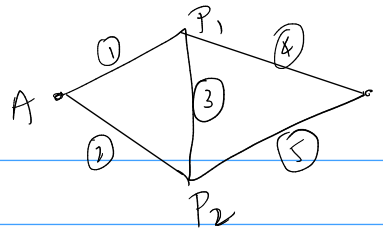
$P(B) = \frac{1}{6} + \frac{4}{6}\cdot\frac{1}{6} + \left(\frac{4}{6}\right)^2 \cdot \frac{1}{6} + \dots = 1/6 \cdot \frac{1}{1 - 4/6} = 1/2$

Condition on first throw

$P(B) = \frac{1}{6} \times 1 + \frac{1}{6} \times 0 + \frac{4}{6} \times P(B)$

$\frac{1}{6}$: first is 4 $\quad \frac{1}{6}$: first is 5 $\quad \frac{4}{6}$: first is 1 or 2 or 3 or 6

(3)



each road open with prob. p indep.

$P(\text{open route from A to B})$

Aug 14

E : Roads $R_1, R_2, \ldots, R_i$ are open & other roads are shut $\quad P(E) = p^i (1-p)^{5-i}$

E_i : exactly i roads are open $\quad i = 0, 1, 2, 3, 4, 5$

$$P(\underbrace{\text{open route from A to B}}_{F}) = \sum_{i=0}^{5} P(F|E_i)\, P(E_i)$$

	$P(F \cap E_i)$	$P(F \mid E_i)$
$i=0$:	0	0
$i=1$:	0	0
$i=2$:	$2p^2(1-p)^3$	$2/\binom{5}{2}$
$i=3$:	$8p^3(1-p)^2$	$8/\binom{5}{3}$
$i=4$:	$5p^4(1-p)$	1
$i=5$:	p^5	1

Coupon collector's problem

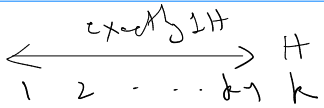
→ n things to collect r different coupons

Ex: Geometric distribution.

Toss a coin $p(H) = p$ repeatedly, indep

$P_r(\text{first heads in } k\text{-th toss}) = (1-p)^{k-1} p \quad k = 1, 2, 3, \ldots$

$P_r(\text{second heads in } k\text{-th toss}) = \binom{k-1}{1} \cdot p^2 (1-p)^{k-2}, \quad k = 2, 3, \ldots$



$$\Pr(r^{th} \text{ head in } k\text{-th toss}) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

$k = r, r+1, \ldots$

$P(A), P(B) > 0$

$$P(A \cap B) = P(A) \cdot P(B) \iff P(A|B) = P(A)$$
$$P(B|A) = P(B)$$

$A_1, \ldots, A_k$: mutually indep.

disjoint $\updownarrow$ $\{i_1, i_2, \ldots, i_s\} \subseteq [k]$, $\{j_1, j_2, \ldots, j_r\} \subseteq [k]$

$$P(A_{i_1}, \ldots, A_{i_s} \mid \underbrace{A_{j_1}, \ldots, A_{j_r}}_{P_r > 0}) = P(A_{i_1}, \ldots, A_{i_s})$$

Bayes Theorem: A, B $\quad P(A), P(B) > 0$

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) P(A)$$

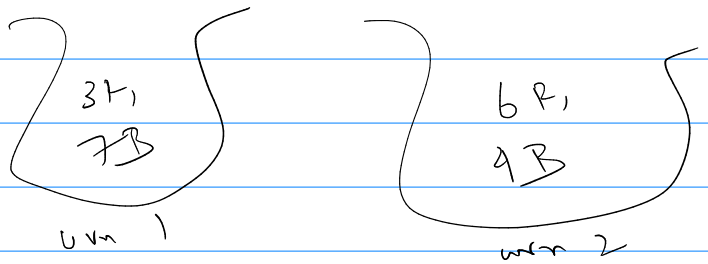
$$P(B|A) = \frac{P(A|B) P(B)}{P(A)}$$

Partition $E_1, E_2, \ldots, E_k$ $\quad P(E_i)$ & $P(A|E_i)$: known

Event A $\quad$ "prior"

"Posterior": $P(E_i|A) = \dfrac{P(A|E_i) \cdot P(E_i)}{\sum_{i=1}^{k} P(A|E_i) P(E_i)}$

Ex: (1)



$$P(\text{urn } 1 \mid R) = \frac{\frac{3}{10} \cdot \frac{1}{2}}{\frac{3}{10}\frac{1}{2} + \frac{6}{10}\frac{1}{2}} = \frac{1}{3}$$

(2) Amazon

prior

$\Pr(p=0) = 1/11$

$p \in \{0, \frac{1}{10}, \frac{2}{10}, \dots, \frac{10}{10}\}$ (each $1/11$)

$\Pr(p=\frac{10}{10}) = \frac{1}{11}$

"Estimation"

5 reviews, 3 liked it

$$\Pr\left(p=\frac{5}{10} \mid \text{data}\right) = \frac{\Pr(\text{data} \mid p=5/10)\cdot \Pr(p=5/10)}{\sum_{k=0}^{10} \Pr(\text{data} \mid p=\frac{k}{10})\cdot \Pr(p=\frac{k}{10})}$$

$$\max_i \frac{\Pr(\text{data} \mid p=i/10)}{\sum_{k=0}^{10} \Pr(\text{data} \mid p=k/10)} \iff \max_i \left(\frac{i}{10}\right)^3 \left(1-\frac{i}{10}\right)^2$$

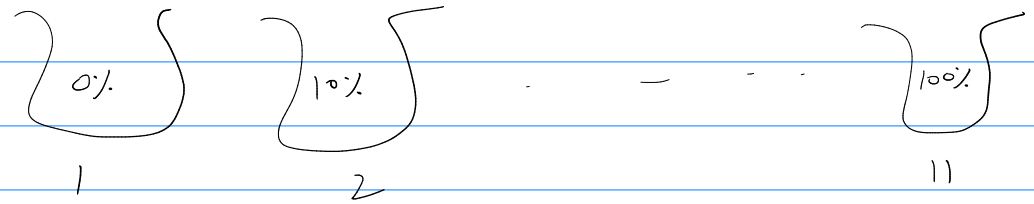
$$\updownarrow$$

$$i^3(10-i)^2$$

$i^* = 6$

(3)

Aug 18



Urn i: $(i-1)10\%$ green, remaining red $\quad p_i = \frac{(i-1)}{10}$

Pick an urn at random. Pick 5 marbles (with replacement) in sequence.

E: last 3 marbles are green, first two are red.

$$P(E \mid \text{Urn } i) = (1-p_i)^2 p_i^3$$

$$P(\text{Urn } i \mid E) = \frac{P(E \mid \text{Urn } i)\cdot P(\text{Urn } i)}{\sum_j P(E \mid \text{Urn } j)\cdot P(\text{Urn } j)} \quad (P(\text{Urn } i) = 1/11)$$

index j

$$\max_i P(\text{Urn } i \mid E) = \max_i P(E \mid \text{Urn } i) \text{ achieved at } i=6$$

(4) Clinical trials

Response \	Imip.	Lit.	Combo.	Placebo	Total
Relapse	18	13	22	24	77
No relapse	22	25	16	10	73
Total	40	38	38	34	150

(5) Disease: affects 1 in 10000. (0.01%)

Test: 90% accurate (positive & negative)

$$P(\text{have disease} \mid \text{Test +ve}) = \frac{1}{1112} \approx 0.1\%.$$

Gambler's ruin

Initial sum: k units

One round: Bet: 1 unit. Win (Pr. p): get 2 units (+1)

Lose (Pr. $q = 1-p$): lose 1 unit (−1)

→ keep gambling till you have N units (cash out) (or) you go bankrupt.



P_k: Prob. of cashing out when starting with k units.

$$P_k = P_r(\text{cash. out} \mid \text{win in 1st round}) \cdot p + P_r(\text{cash. out} \mid \text{loss in 1st round}) \cdot q$$

P_{k+1} $\qquad$ P_{k-1}

$$pP_k + qP_k = P_k = pP_{k+1} + qP_{k-1} \qquad P_0 = 0, \quad P_N = 1$$

$$q(P_k - P_{k-1}) = p(P_{k+1} - P_k) \qquad (p > 0)$$

$$P_{k+1} - P_k = \frac{q}{p}(P_k - P_{k-1})$$

$$= \left(\frac{q}{p}\right)^2 (P_{k-1} - P_{k-2}) = \cdots = \left(\frac{q}{p}\right)^k P_1$$

Telescoping. $$P_k = \left(\left(\frac{q}{p}\right)^{k-1} + \cdots + \frac{q}{p} + 1\right) P_1$$

$k = N$ $$P_N = \left(\left(\frac{q}{p}\right)^{N-1} + \cdots + \frac{q}{p} + 1\right) P_1 = 1$$

$p = q$: $\quad P_1 = 1/N \Rightarrow P_k = k/N$

$p \neq q$: $$P_1 = \frac{\frac{q}{p} - 1}{\left(\frac{q}{p}\right)^N - 1} \Rightarrow P_k = \frac{\left(\frac{q}{p}\right)^k - 1}{\left(\frac{q}{p}\right)^N - 1}$$

→ Chap 1 & Chap 2 Tut 1 + 2

Quiz 1 → Aug 25, 7:45 am

5 Qns, 5 marks each

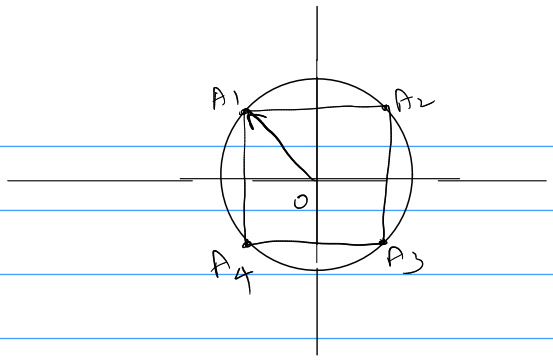
1 A4 handwritten sheet

Probabilistic method

Sphere → surface painted red + blue (10%)

(same centre)

→ is there an inscribed cube all of whose vertices are red?



$rot(A_i)$: blue

Graphs

A, B, C, D : graduates

p: any two become friends, indep.

A: hears rumour $Pr(D \text{ will hear it}) = ?$

A • • B

Random Walk

D • • C

Cards Bridge

13 cards

4S 3H 3C 3D

vs.

4S 4H 3C 2D

$\binom{13}{3}^2$ vs. $\binom{13}{4}\binom{13}{2}$

$\frac{13^2 \times 12^2 \times 11^2}{6^2 6^2}$ vs. $\frac{13 \times 12 \times 11 \times 10}{24} \times \frac{13 \times 12}{2}$

4333 vs. 4432

$\frac{\binom{13}{4}\binom{13}{3}^3}{\binom{52}{13}}$ > $\frac{\binom{13}{4}^2\binom{13}{3}\binom{13}{2}}{\binom{52}{13}}$